

IPS II

Interference Pattern Structured Illumination Imaging

→ Way to capture Images:

→ No lens.

→ create multi-pixel image with single pixel detector

→ applicable to any type of wave.

(light - x-ray - radar - acoustic waves - quantum waves)

→ Goal to overcome limits of conventional imaging:

→ Resolution [wavelength of light & size of optic]

→ Working distance.

(need either short distance from object, or big lens to capture big angle coming from the light)

→ Depth of field (range of things which are in focus)

low Res. → large range

high Res. → short range

→ Field of view.

for high Res. need large numerical aperture
($n \sin \theta$)

Resolution Limit

$$d = \frac{\lambda}{2 n \sin \theta}$$

→ lenses / mirror use multi pixel detector.

* How IPSII Works:

"Structured illumination Imaging"

Rather than making image onto Multi-Pixel detector,
we control shape of light that's illuminating the object
to gain info. about the object

Simply by measuring how light scatters from that pattern

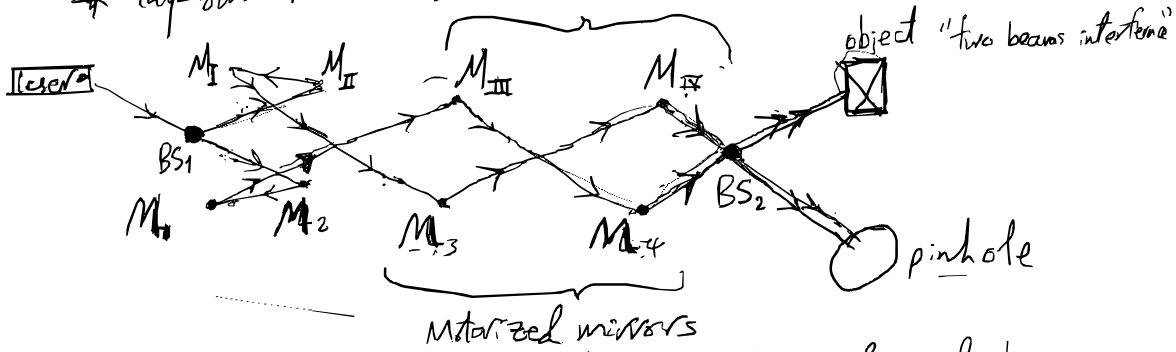
~~•~~ "IDAR" → use structure illumination.

~~•~~ "IPSII" → use Interference Pattern structural illumination

* Advantages of IPSII:

- No high "numerical aperture" lens/mirror needed
- No need for being close to object.
- Depth of field & field of view
- All waves (x-ray, electron, radar, far IR, acoustic)
- Compressed sensing:
 - 2x resolution
 - similar to MRI tech.

* lay-out of ZPSII: "Interferometer"



* we control angle and position of each beam

* (Relative Phase btwn two laserbeams)

determine dark and bright stripes in interference pattern

* Future work:

Experimental laser optics

Mechanical design and testing

Data processing methods

Image processing

MRI parallel imaging techniques.

Computer motion control.

Digital Holography

Ultra Sonic acoustic imaging

High-Res Long-working-distance microscopes.

Boosting imaging speed.

Radar
"no big dish"

AOM